Volvo Trucks - CH3 SVC

Camera Interface Specification

Version 1.0

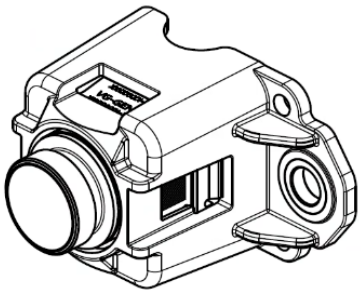
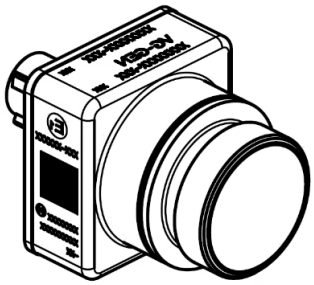
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Name of Customer:	Volvo Trucks	
Bosch Product Name*	CH3 SVC incl. Bracket	CH3 SVC
Volvo Trucks Product Name*	Near Range Camera incl. Bracket	Near Range Camera
Mounting location	1. Front apron 2. Side mirror / side wing	Rear view shutter module
Connector	FAKRA	FAKRA
C1-sample		
Customer Part Number	C1-A: 24569898.B03 C1-B: 24569898.C04	24918084.C02
Customer Part Name	C1-A- / C1-B-sample	C1-sample
Bosch Part Number	0275B02127-01	0275B02128-01
Offer drawing number	0273A02229	0273A02239
Product Picture		
Drawing Version	06	01

*: from section 1 onwards, component is just called camera

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1. Scope

The purpose of this document is to inform about the Camera Interface in detail.

This document is aimed to be shared with Volvo Trucks. Sending the document to other third-party component suppliers is not allowed. Exceptions need to be agreed by Robert Bosch GmbH.

This Camera Interface Specification is valid for the following CH3 SVC samples:

C-sample		
Component	Manufacturer	Version Major
Image Sensor	Samsung	S5K3B6SAA3-Z172-KA2KAV
Serializer	Analog Devices / Maxim	MAX96717FGTJ/VY+T
PMIC	Rohm	BD868W0MUH-C
EEPROM	Rohm	BR24H32
I2C-Settings		
Serializer	SER-Setting-Name: • 60 fps: SVC3B6_SerDes_VolvoTruck_SXPF_I2C_**Pixel_RAW12**_C1_20251107 • 30 fps: SVC3B6_SerDes_VolvoTruck_SXPF_I2C_**Pixel_RAW16**_C1_20251107	
Image Sensor	**C1-sample** settings: • IMG-Setting-Name (application relevant HDR setting): ○ 60 fps: "S5K3B6SA_MAXIM_HDR4_RSSR_12B_MCLK25_60fps_1920x1550_MIPI4_630Mbps_wGamma_wEMBD_wSTAT6line_wFuSa_wSECU_wDS_20251029.sset" ○ 30 fps: "S5K3B6SA_MAXIM_HDR4_RSSR_16B_MCLK25_30fps_1920x1550_MIPI4_481.25Mbps_wGamma_wEMBD_wSTAT6line_wFuSa_wSECU_wDS_20251029.sset" • Originator of settings: Samsung • Embedded data: ON **B2-sample** settings (compatible with C-sample HW): • Please refer to B2-sample settings information	
Functional Safety Features		
Image Sensor	**C1-sample** Settings: • FuSa-configuration included in I2C-IMG-setting • Features: Volvo Trucks specific • I2C CRC: ON **B2-sample** settings (compatible with C-sample HW): • Please refer to B2-sample information	
PMIC	• Failure reactions: Volvo Trucks specific • I2C CRC: ON	

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Cybersecurity Features	
Key fusing (Image Sensor related)	Yes
Device Authentivication (Image Sensor related)	Enabled (via IMG key. Function hosted by ECU)
Image Authentification (Image Sensor related)	Enabled (via IMG key + I2C config. Function hosted by ECU)
EEPROM authen-ticity and integrity (EEPROM related)	Enabled (via EEPROM key. Function hosted by ECU)

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2. Start-up Procedure

For start streaming of the camera, the following steps need to be considered from host (e.g. ECU):

1. Enable POC i.e. power up voltage supply
2. Host to write I2C configuration of SerDes (SerDes setting)
3. Host to write I2C configuration of imager (imager sensor setting)
4. Host to provide sync-signal for image sensor (refer to chapter 9 and 10)

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3. I2C addresses

Component	I2C Address [8-Bit]
Image Sensor	0x20
Serializer	0x80
PMIC	0xE0
EEPROM	0xA0

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4. MIPI D-PHY Routing

Image Sensor Output		Serializer Input	
Lane	Polarity	Lane	Polarity
CLK	N	CLK	P
	P		N
0	N	1	P
	P		N
1	N	0	P
	P		N
2	N	2	P
	P		N
3	N	3	N
	P		P

Register setting of Serializer to accommodate lane and polarity swaps [Slave Address (8-Bit), Register Address (16-Bit), Register Value (8-Bit)]:

- 0x80, 0x0332, 0xE0
- 0x80, 0x0333, 0x01
- 0x80, 0x0334, 0x10
- 0x80, 0x0335, 0x07

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5. Serializer Configuration

B1-sample and B2-sample					
MFP[1]	**Name of net**	**Description of net**	**External Pulls**	**Default state (camera powered, no config of camera)**	**Target Configuration of MFP**
0	IMG_RST_N	Enable (high) or disable (low) imager	Pull down	Low	Push-pull high or Low
1	CFG0_FSYNC	Strap during start-up; Frame Sync connection to imager	Strap set to be I2C, XTAL, 0x80		Push-pull high or Low
2	CFG1	Strap during start-up;	Strap set to be COAX, Pixel, 3Gbps		
3	SER_LH_EN	Enable (high) or disable (low) lens heating	Pull down	Low	Push-pull high or Low
4	SER_CLKOUT	Programmable clock output of the serializer; Used as imager clock input		Low	25MHz
5	PMIC_ERROR_N	Error input from PMIC; No error (high), error (low)		High or low	Input internal pull down
6	VARIANT_CODE	Voltage divider input: low: B1-Sample (w/ LH)	Pull down	Low	Input No internal pull
7	IMG_ERROR_N	Error input from imager; No error (high), error (low)	Pull down	High or low	Input internal pull down
8	EEPROM_WP	Write protect of EEPROM: high: protected low: write enabled	Pull up	High	Push-pull high or Low

[1] MFP: Multi-Functional PIN

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C1-sample and C2-sample					
MFP[2]	**Name of net**	**Description of net**	**External Pulls**	**Default state (camera powered, no config of camera)**	**Target Configuration of MFP**
0	IMG_RST_N	Enable (high) or disable (low) imager	Pull down	Low	Push-pull high or Low
1	CFG0_FSYNC	Strap during start-up; Frame Sync connection to imager	Strap set to be I2C, XTAL, 0x80		Push-pull high or Low
2	CFG1	Strap during start-up;	Strap set to be COAX, Pixel, 3Gbps		
3	SER_LH_EN	Enable (high) or disable (low) lens heating	Pull down	Low	Push-pull high or Low
4	SER_CLKOUT	Programmable clock output of the serializer; Used as imager clock input		Low	25MHz
5	PMIC_ERROR_N	Error input from PMIC; No error (high), error (low)		High or low	Input internal pull down
6	LH_Readback	Digital readback of lens heating voltage: lens heating enabled (high) lens heating disabled (low)	Voltage divider to meet digital IO levels	Low	Input No internal pull
7	IMG_ERROR_N	Error input from imager; No error (high), error (low)	Pull down	High or low	Input internal pull down
8	EEPROM_WP	Write protect of EEPROM: high: protected low: write enabled	Pull up	High	Push-pull high or Low

[2] MFP: Multi-Functional PIN

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6. EEPROM

6.1. EEPROM General

This document refers to this Bosch EEPROM layout (refer also to description of general header block):

- EEPROM Layout version = Version 1.0, 2024-07-24
- EEPROM Customer = 0x05
- EEPROM Customer Variant Flag = 0x00
- VersionMajor = 0x00
- VersionMinor = 0x00

EEPROM layout history:

EEPROM layout version	Version Major	Version Minor	Changes
Version 1.0	0x00	0x00	Baseline

EEPROM layout camera history:

Sample	Version Major	Version Minor	Remark
CH3 SVC B1-sample	0x00	0x00	
CH3 SVC B2-sample	0x00	0x00	
CH3 SVC C1-sample	0x00	0x00	

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6.2. EEPROM Memory

This section describes EEPROM memory of camera.

Figure 1 below describes block diagram of CH3 SVC camera excluding system integration.

Camera contains EEPROM memory.

Size of EEPROM memory is 32 kBit.

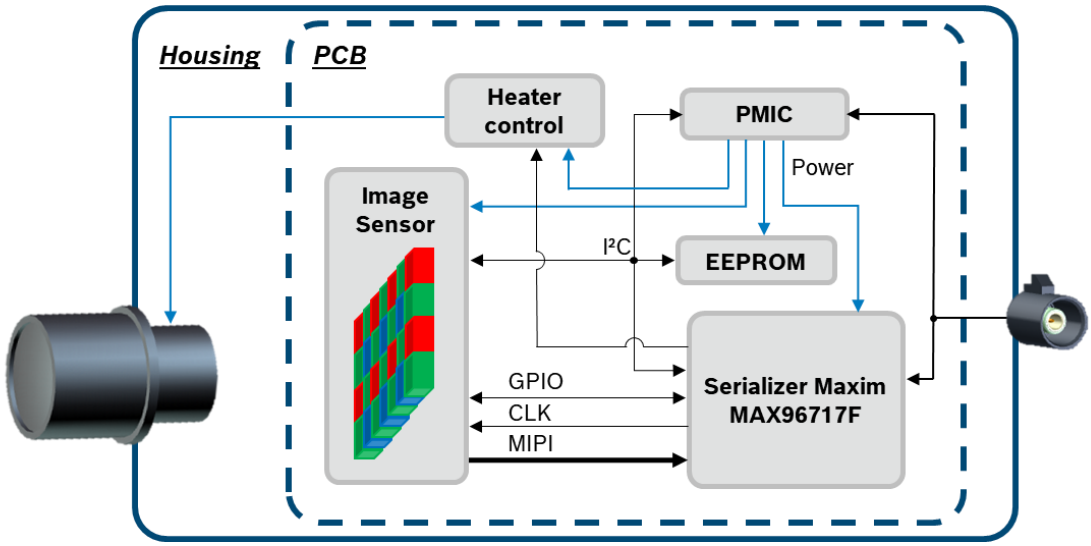


Figure 1: Camera-level block diagram

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6.3. EEPROM MAP

6.3.1. Layout

Figure 2 below describes basic overview of EEPROM layout. All blocks are mentioned including size and planned access. Read-Only in column *Access by ECU* does not mean that write protection is implemented but does mean that this block shall not be written by ECU. Column *Write access* describes who shall do write access to this block.

No.	Block Name	Size w/ optional data	Size w/o optional data	Operatio-nal data	Access by ECU	Write access
1	General Header Block	32	32	yes	Read-Only	Bosch plant
2	Label Data Block	224	224	yes	Read-Only	Bosch plant
3	Intrinsic Data Block	352	352	yes	Read-Only	Bosch plant
4	Block for Bosch plant data	64	64	yes	Read-Only	Bosch plant
5	Security (RO Data) Block	96	96	yes	Read-Only	Bosch plant
6	OEM Data Block	64	64	no	Read-Write	ECU
7	Optional: Failure Entry block	448		no	Read-Write	ECU
8	Optional: Temperature Histogr. Block	96		no	Read-Write	ECU
	used space [Bytes]	1376	832			
	free space [Bytes]	2719	3263			
	degree of filling	33.6%	20.3%			

Figure 2: EEPROM layout overiew (1)

Figure 3 below describes more details of the blocks of the EEPROM layout. ID, name, start address, size, basic contents and use of signature and security per block are described.

#ID	Block Name	Start-address	Size	Block description	Signature (Security) / CRC (Safety)
0x01	General Header Block	0x0000	32	General information about the EEPROM	CRC for each data block
0x02	Label Data Block	0x0020	224	Label data of the camera for traceability	
0x04	Intrinsic Data Block F-Theta model	0x0100	352	Determined parameters during intrinsic	
0x05	Internal Part Information Block	0x0260	64	Placeholder for internal part information	
0x0F	Security (Bosch RO Data) Block	0x02A0	96	Contains security signature for all read only blocks above written in	ECDSA-Signatur for RO data (RO blocks above) & CRC.
0x10	OEM Data Block	0x0300	64	Contains specific identifier data written by OEM	CMAC-Signature and CRC for each data block CRC is not part of CMAC calculation, but CRC includes CMAC value (CRC secures signature)
0x11	Failure Entries Block	0x0340	832	Failure entries defined by Bosch & customer	
0x12	Temperature Histogram Block	0x0680	96	Histogram of operating temperature of	

Figure 3: EEPROM layout overiew (2)

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6.3.2. Data storage

Data stored in "Big-Endian" Format.

Data format of intrinsic data is "32-bit single precision" or "64-bit double precision" according to IEEE754 (refer to description of parameters for detailed information whether parameter is single or double precision).

6.3.3. Block structure

EEPROM layout is organized in blocks. Each block has a certain size and CRC checksum is calculated and stored per block.

Data in blocks consists of several elements. Each element contains some data.

Address offset of element is relative to start address of block.

Size represents size in bytes of data type. Field counter describes how many fields of data type are stored in the lement. Length describes length of element.

Example: An element might contain 6 fields of unit16. Size is 2 bytes because unit16 is 2 bytes. Field counter is 6 because element contains 6 fields. Length of element is 2 x 6 = 12 bytes.

This is data stored in first three elements of each block.

Address offset	Data type	Size [byte]	Field counter	Length [byte]	Parameter Name	Remark
0	uint8	1	1	1	block type ID	
1	unit16	2	1	2	block length	Total number of bytes used for block
3	unit24	3	1	3	block CRC write counter	Incremented at each write of CRC

This is data stored in last element of each block.

Address offset	Data type	Size [byte]	Field counter	Length [byte]	Parameter Name	Remark
Block length -2	uint16	2	1	2	CRC checksum	Checksum calculated over all fields of the Block except crc

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6.3.4. Calculation of CRC checksum

This is CRC Algo for individual data blocks: CRC-16/MCRF4XX.

- Polynomial: 0x1021
- Initial value for CRC calculation: 0xFFFF
- Final Xor Value (XorOut): 0x0000
- Input Reflected (RefIn) / Output Reflected (RefOut) = true
- Example: String "Hello World!"
    - Converted to Hex Format: 0x48 0x65 0x6C 0x6C 0x6F 0x20 0x57 0x6F 0x72 0x6C 0x64 0x21 -> Resulting CRC: 0xF444

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6.3.5. General header block

Table below lists the information stored in the general header block. Length of general header block is 32.

Address offset [dec]	Data type	Size [byte]	Field counter	Length [byte]	Parameter Name	Remark
0	uint8	1	1	1	block type ID	0x01
1	unit16	2	1	2	block length	32
3	unit24	3	1	3	block CRC write counter	
6	unit32	4	1	4	...	
10	uint8	1	1	1	EEPROM Customer Flag	Volvo Trucks = 0x05
11	uint8	1	1	1	EEPROM Customer Variant Flag	SVC = 0x00 (F-Theta intrinsic block)
12	uint8	1	1	1	Major version	0x00
13	uint8	1	1	1	Minor version	0x00
14	uint8	1	1	1	...	
15	unit8	1	15	15	...	
30	uint16	2	1	2	CRC checksum	

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6.3.6. Label data block

Table below lists the information about the label information stored in the label data block. Length of label data block is 224.

Address offset [dec]	Data type	Size [byte]	Field counter	Length [byte]	Parameter Name	Remark
0	uint8	1	1	1	block type ID	0x02
1	unit16	2	1	2	block length	224
3	unit24	3	1	3	block CRC write counter	
…						
8	unit8	1	16	16	Imager-ID	
24	unit8	1	1	1	Imager Revision	
25	string	1	10	10	Bosch part number	written on label
…						
45	string	1	3	3	Index of Bosch P/N	written on label
…						
64	string	1	3	3	Bosch plant code	Blaichach is “050”
67	string	1	8	8	Production date “yyyymmdd”	Part of unique serial number
…						
83	string	1	4	4	Line on which camera was produced	Part of unique serial number
87	unit32	4	1	4	counts cameras pro-duced per day	Part of unique serial number
…						
111	string	1	24	24	Lens DMC contents	
…						

See continuation on next page.

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Continuation:

Address offset [dec]	Data type	Size [byte]	Field counter	Length [byte]	Parameter Name	Remark
135	string	1	6	6	ID of OEM	World manufacturing code
141	string	1	20	20	OEM part number (20 digits)	Written on label. • Bracket variant: bracket assembly part number • CAM-only variant: camera part number
170	string	1	30	30	OEM Serial No	Currently not used.
...						
222	uint16	2	1	2	CRC checksum	

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6.3.7. Intrinsic data block (F-Theta)

Table below lists the information about the intrinsic parameter stored in the intrinsic data block. Length of label data block is 352. Model is F-Theta.

Address offset [dec]	Data type	Size [byte]	Field counter	Length [byte]	Parameter Name	Remark
0	uint8	1	1	1	block type ID	0x04
1	unit16	2	1	2	block length	352
3	unit24	3	1	3	block CRC write counter	
...						
8	unit16	2	2	4	Calibration Model Version	0x0001, 0x0001 (F Theta 1.1)
12	unit8	1	1	1	Intrinsic Calibration Model	0x0A (F Theta)
13	string	1	8	8	Bosch software version	
21	uint16	2	1	2	Image rows	Number of rows of image the intrinsic is referenced to (full sensor read-out ROI)
23	uint16	2	1	2	Image columns	Number of columns of image the intrinsic is referenced to (full sensor read-out ROI)
25	uint16	2	1	2	Start Row	Start Row of ROI, End Row = Imager Rows - Start Row
27	uint16	2	1	2	Start column	Start Column of ROI, End Column = Image Columns - Start Column
29	float32	4	1	4	Camera constant along y-direction	[pix]
33	float32	4	1	4	Aspect ratio	
37	float32	4	1	4	Principal point x	[pix]
41	float32	4	1	4	Principal point y	[pix]
45	float32	4	12	48	Distortion parameters for transformation from undistorted to distorted image (U2D)	= [A1, A2, A3, A4, A5, A6, A7, A8, B1, B2, C2, r0]
93	float32	4	12	48	Undistortion parameters for transformation from distorted to undistorted image (D2U)	= [A1, A2, A3, A4, A5, A6, A7, A8, B1, B2, C2, r0]

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...						
141	Float32	4	3	12	Internal Angles	Orientation of the camera wrt. Nominal orientation in the fixture coordinate-system in [°] defined in DIN70k-coord. Sys. = [Roll, Pitch, Yaw].
...						
231	float32	4	1	4	Valid FOV	Field of View within which the intrinsic is valid in [°]
235	float32	4	1	4	Valid Image Radius	Image-Radius within which the intrinsic is valid in [pix]
...						
350	uint16	2	1	2	CRC checksum	

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7. Lens Heating Control

7.1. Disclaimer

- Please refer to the conditions provided TCD[3] (Document Number: 0220DN00E0).
- Information to be used as reference only. Depends on specific use-case.

7.2. Activation and monitoring of lens heating

Please refer to:

- MFP controls in Serializer Configuration in section 5.
- Description contained in the TCD (reference see above).

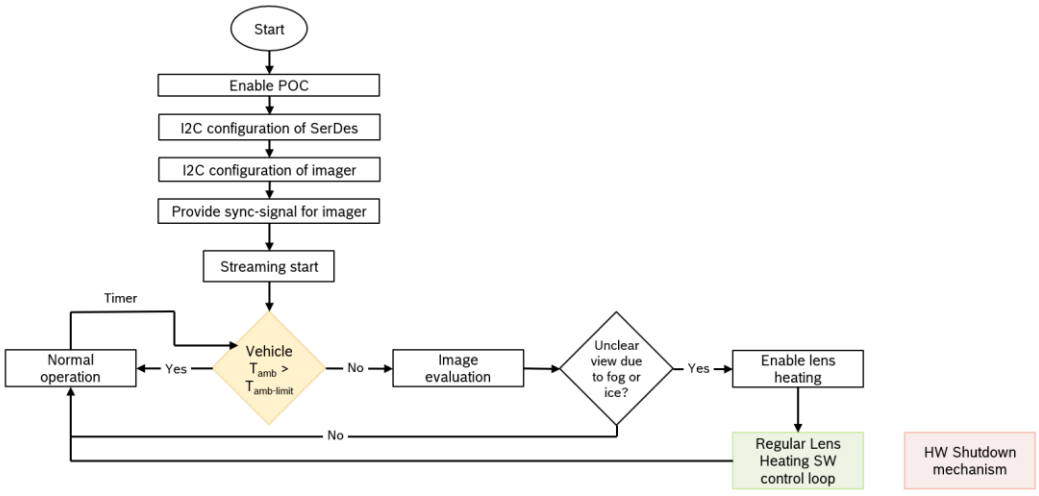
7.3. Lens heating software control loop

The software control loop shall be integrated via multiple stage process which includes:

1. Evaluation of image data
2. Ambient temperature T_{amb}
3. Image sensor junction temperature $T_{J,IMG}$
4. Serializer junction temperature $T_{J,SER}$
5. Timer t

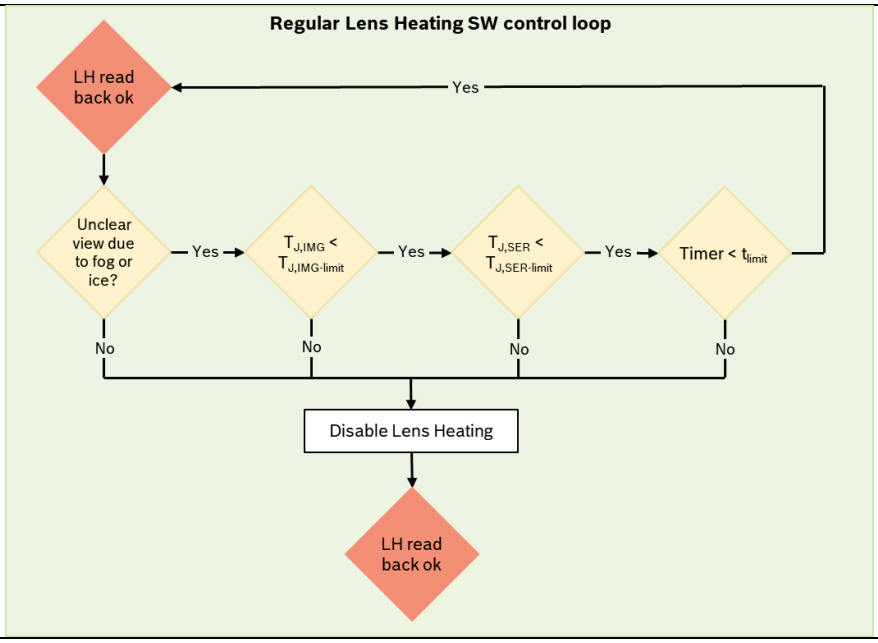
The frequency of reading out the upper mentioned temperatures shall be >= 1 Hz.

7.3.1. Description of software control loops

	Flow Chart
Loop 1 Process steps prior to enter the "Regular Lens Heating SW control loop"	

[3] TCD: Technical Customer Documentation

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Loop 2 "Regular Lens Heating SW control loop"	

Pre-conditions for activation

Based on B-sample validation the following pre-conditions for activation are recommended:

1. Evaluation of image data:
   The necessity of switching ON the LH-function is verified (see diagram "Unclear view due to fog or ice?")
2. Temperature limits:
   - $T_{amb\text{-}limit}$ = 10°C
   - $T_{J,IMG\text{-}limit}$ = 45°C
   - $T_{J,SER\text{-}limit}$ = 40°C
3. Timer limit:
   - t_{limit} = 5 min

The upper mentioned temperature limits must be verified on vehicle level and depend on the thermal properties of the installation space.

Switching cycles

The camera HW is designed to ensure 10,000 ON/OFF switching cycles with a cycle time of $t_{LH,cyc} \geq 1s$ i.e. $f_{LH} \leq 1Hz$.

Installation Space

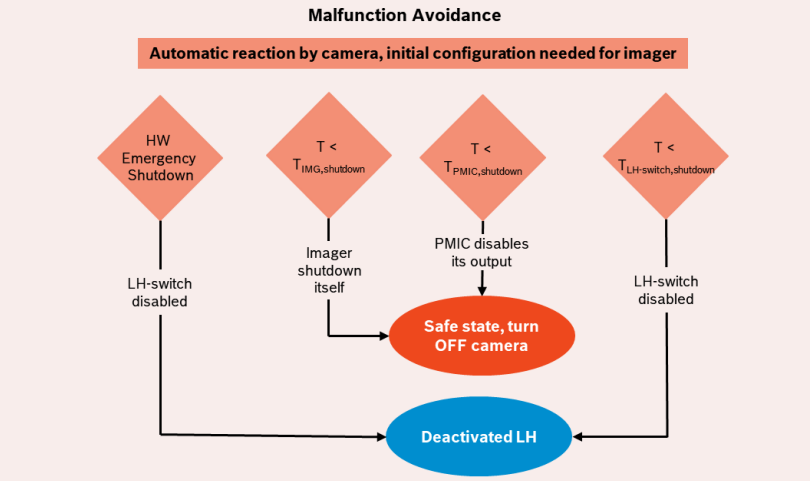
A temperature limit for the steady state (streaming ON, lens heating OFF, $t_{meas} \geq t_{steady_state}$) which characterizes the **installation space** shall be kept:

$$\Delta T = T_{J,IMG} - T_{amb} \leq 20\ K$$

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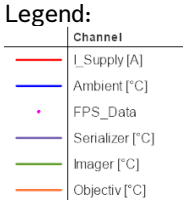
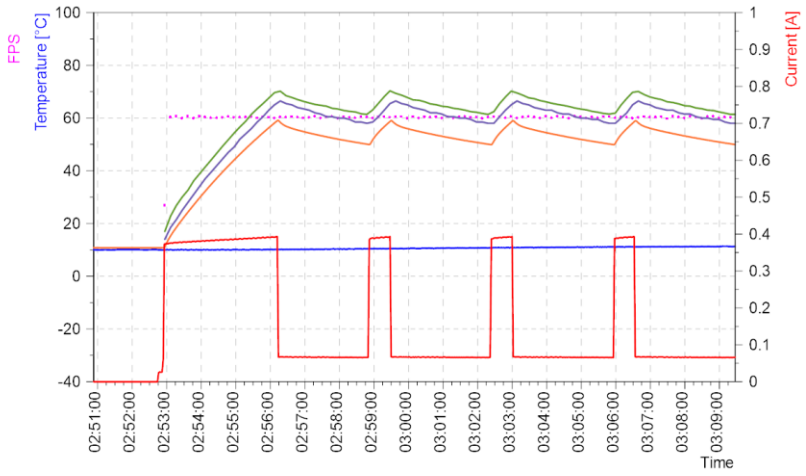
7.4. Camera Hardware Emergency Shutdown

- In order to prevent overheating of the camera front lens resulting in hot surface hazard for humans an emergency shutdown is implemented. The shutdown will take place at ≤ 60 °C lens temperature.
- The emergency shutdown functionality is integrated via HW and triggered automatically by the camera itself. There is no action by the host required.
- The software control loop shall stop LH-function before emergency shutdown becomes effective.
- In case of an emergency shutdown, after cooldown of the camera HW, the LH-circuit is enabled and responsibility shifts back to SW.
- Detailed Description of the HW shutdown functional block and its implementation:

<table>
<tr><th>Description</th><th>Flow Chart / Plot</th></tr>
<tr><td>HW emergency shut-down mechanism</td><td> 

Shutdown temperatures:
• $T_{IMG,shutdown}$ = 125 °C
• $T_{PMIC,shutdown}$ = 140 °C
• $T_{LH\text{-}Switch,shutdown}$ = 160 °C</td></tr>
<tr><td>Conditions:
• Camera in Bracket
• At t = 0s CAM starts streaming and LH is activated.

Plot description:
• LH functionality is stopped by tempera-ture limitation of the NTC several times when reaching the limit.

Legend:
 
</td><td> 
</td></tr>
</table>

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8. Temperature Sensors

The camera features multiple temperature sensors:

Device	Description	Temperature Threshold	Accuracy
PMIC	Thermal Shutdown	140 °C	+/- 10 °C
Lens Heating Switch	Thermal Shutdown	160 °C (30 °C Hysteresis)	n/a
Serializer	Temperature Sensor 1	-	-40 °C – 105 °C: +/- 3.0 K 105 °C – 125 °C: +/-2.5 K
	Temperature Sensor 2		
Image Sensor	Temperature Sensor 1	See Chapter 10	-40 °C – 105 °C: +/- 3.0 K 105 °C – 115 °C: +/- 2.0 K 115 °C – 125 °C: +/- 1.0 K
	Temperature Sensor 2		

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9. Image Sensor Settings

9.1. General

The following summary provides an overview of the Samsung 3B6 image sensor setting which was used in Design Validation.
During product development this setting might experience changes.

9.2. Function Overview

B1-sample and B2-sample		
Function	**Setting 1**	**Setting 2**
Short Description	**60 fps** setting (DV)	**30 fps** setting
General Parameters		
Resolution	1920 px x 1536 px (max.)	←
Bit depth per Pixel	12 bit	16 bit
Frame rate	60 fps	30 fps
INCK	25 MHz	←
Sync Mode	OFF	←
Gains	Use recommended gains from supplier.	←
Exposure time	11 ms (adaptable)	←
Pedestal	168	←
MIPI non-continuous mode	OFF	←
Frame Format		
Image orientation	No Flip, No Mirror	←
Optical center	No decentering	←
Embedded data (Frame Format)	Activated, fixed	←
Data type of embedded data	same type as image (0x2C)	same type as image (0x2E)
Statistics	OFF	←
Test Pattern (Safety)	OFF	←
Frame Sequence	No context switching, VC0	←
HDR Frame		
Frame Mode	HDR4	←
Apply Factor	ON	←
Piecewise linearization	Recommended setting from supplier used.	←
HDR merger configuration	Use recommended setting from supplier.	←
Phase correction	OFF	←
Post Processing		
Static Bad Pixel Correction	OFF	←
Dynamic Bad Pixel Correction	OFF	←
Noise Reduction	OFF	←
White Balance	OFF	←
Auto Exposure	OFF	←
Functional Safety		
Configuration of Safety Measures	Platform configuration	←
I2C CRC	OFF	←
Cyber Security		
Image Authentification by AES-CMAC	None	←
Decryption waiting time	n/a	←
AES-CMAC input data	n/a	←

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C1-sample		
Function	**Setting 1**	**Setting 2**
Short Description	**60 fps** setting (DV)	**30 fps** setting
General Parameters		
Resolution	1920 px x 1536 px (max.)	←
Bit depth per Pixel	12 bit	16 bit
Frame rate	60 fps	30 fps
INCK	25 MHz	←
Sync Mode	Optional. External trigger FSYNC high signal ≥ 1.5µs	←
Gains	Use recommended gains from supplier.	←
Exposure time	11 ms (adaptable)	←
Pedestal	160	←
MIPI non-continuous mode	ON	←
Frame Format		
Image orientation	No Flip, No Mirror	←
Optical center	No decentering	←
Embedded data (Frame Format)	Activated, fixed	←
Data type of embedded data	same type as image (0x2C)	same type as image (0x2E)
Statistics	ON	←
Test Pattern (Safety)	ON	←
Frame Sequence	No context switching, VC0	←
HDR Frame		
Frame Mode	HDR4	←
Apply Factor	ON	←
Piecewise linearization	Recommended setting from supplier used.	←
HDR merger configuration	Use recommended setting from supplier.	←
Phase correction	OFF	←
Post Processing		
Static Bad Pixel Correction	ON	←
Dynamic Bad Pixel Correction	OFF	←
Noise Reduction	OFF	←
White Balance	OFF	←
Auto Exposure	OFF	←
Functional Safety		
Configuration of Safety Measures	VT-specific configuration	←
I2C CRC	ON	←
Cyber Security		
Image Authentification by AES-CMAC	Configured (calculation based on key written in OTP)	←
Decryption waiting time	Excluded. Fast start-up (decryption available after 17th frame onwards)	←
AES-CMAC input data	Header & Image Data	←

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9.3. Error Flags Configuration

- Error pin is set to “active low” i.e. low in case of an error.
- Error flags value is per default “1” and it will switch to “0” in case of an error. (valid for error bits in registers and embedded data)

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10. Functional Safety Features

The Camera Head implements the functional safety features, some examples:

- Voltage and temperature monitoring
- Monitoring for internal hardware faults
- Provide data for the system level safety mechanism (Image CRC, Frame Rate, Error Flags)

For details see the "*TSC*" and "*Requirement from Camera Head to HOST SYSTEM*". See references in this Chapter

Image Synchronization

The application will use the signal "FRAME_SYNC" to synchronize the mage capturing in the Camera Heads.
A safety mechanism implemented in the Camera Head to monitor the toggling of FRAME_SYNC signal in each frame and detect faults (i.e.: signal stuck at LOW). In case of failure transition to safe state is triggered. Applied safe state:

- Safe State 01: video stream stopped, GMSL Link alive, communication is possible with Camera Head

Note: *If the FRAME_SYNC is NOT generated by the HOST SYSTEM for the Camera Head, it is considered "stuck at LOW" failure and the Camera Head will stop the video stream. The test setups should be prepared to generate FRAME_SYNC*

References

1. ***Technical Safety concept (TSC)***
   - Title: "Technical Safety Concept of VT CH3 SVC (Camera Head)"
   - Description: Technical Safety Concept of Camera Heads
   - Format: PDF
   - Revision: 17.2 (2025-09-15)
2. ***Requirement from Camera Head to HOST SYSTEM***
   - Title: "Functional Safety Requirement for HOST SYSTEM"
   - Description: Functional Safety Requirements from Camera Head against to HOST SYSTEM
   - Format: PDF (8 pages)
   - Revision: 17.2 (2025-10-22)

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11. Security Features

Imager Security Features

- Every imager comes with a *device specific RSA key pair and a device* ID written into the Imager OTP.
- The *device specific RSA public key* along with the *Device ID* of every imager is signed by the *Root Private Key* from Samsung. The Root Private Key is stored securely in the Samsung KMS and is not externally accessible.
- The *Root Public Key (related to the Root Pivate Key mentioned above)* from Samsung is shared to Bosch and Bosch has already shared this with Volvo.
- The Root Public Key should be used by the Host ECU to verify the authenticity of the Imager by verifying the signature computed over the *device specific RSA public key* along with the *Device ID*. The Host should read the *device specific RSA public key + Device ID + Signature* from the Imager OTP via the I2C to perform the verification.
- After successful verification, the ECU shall then generate a random session key along with the row select information, encrypt it with the *device specific RSA public key* and send it back to the Imager via the I2C.
- The Imager will then decrypt the session key and the row information using the *device specific RSA private key* and will proceed to use the session key and the row information to compute the AES CMAC.
- The AES CMAC will be embedded in the footer of the image frame and will be sent out to the host ECU via the MIPI.
- The host ECU will compute the AES CMAC from the image frame and compare it to the AES CMAC received on the image frame.
- The Host ECU should use the Header and Image data as input for the computation of the AES-CMAC by selecting appropriate values for the Start H, Interval H and Total H. The start H should preferably be 0 to include the frame counter included in the Header.
- The CMAC computation is only available from the 18th frame to speed up the camera startup. This means that the first 17 frames are without the AES-CMAC Tag.

EEPROM Security Features

- The Read Only sections on the EEPROM are secured for integrity and authenticity.
- A signature is generated on the Read Only data of the EEPROM in the Bosch plant and is then written into the Signature section (refer EEPROM memory layout) of the EEPROM.
- Bosch has already shared a Bosch public key with Volvo. This Bosch public key should be used to verify the signature and thus authenticate the RO data on the EEPROM.
- The Host ECU should always verify the integrity and authenticity of the EEPROM data before using it.

References

- Imager Security concept is aligned in the document “**S5K3B6SA_Cybersecurity_application_note_Rev0.27.pdf**”
- Samsung Key Provisioning is aligned in the documents “**Security_Policy_REV1.00.pdf**” and “**CYBERSECURITY_KEY.FUSING_GUIDE.pptx**”
- EEPROM Security concept is aligned in the document “**EEPROMSecurityConceptForFutureCameraHead.pptx**” stored in the Volvo Sharepoint

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12. Revison

This section describes revision of the document.

Revision	Date	Author	Status	Description
0.1	2025-06-03	Marc Schmid	Draft	• Initial revision, incomplete
0.2	2025-10-13	Marc Schmid, Annika Wilms	Draft	• Document updated for C1-sample
0.3	2025-10-24	Marc Schmid	Draft	• Document corrected and updated for C1-sample
0.4	2025-11-03	Marc Schmid	Draft	• Document corrected and updated for C1-sample
1.0	2025-11-10	Marc Schmid, Thorsten Huber, Tibor Racz, Ravi Ananthaneni	Draft	• Document updated for C1-sample.

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